FINAL EXAM NETWORK COMMUNICATIONS SESI 2012/2013 SEM 2

PART A [30 MARKS]

1. DHCP server provides the following information to its client **EXCEPT**:
   A. IP address
   B. Subnet mask
   C. Maximum Transfer Unit (MTU)
   D. Default gateway

2. All the following statements are true about Internet Control Message Protocol (ICMP) **EXCEPT**:
   A. ICMP is used for echo request/reply (used by ping)
   B. ICMP is used for Error correction
   C. ICMP uses UDP at transport layer.
   D. ICMP is used for Error reporting.

3. In broadcasting algorithms, recipient addresses are determined by using the following methods **EXCEPT**:
   A. Flooding:
   B. Controlled flooding
   C. Fixed tree
   D. Spanning tree

4. Subnet is used for the following purposes **EXCEPT**:
   A. To extend the network ID
   B. To shrink the Host ID
   C. To divide a given IP into sub groups of IP
   D. To make classful addressing scheme better than classless.

5. What type of service model is used by the Internet network architecture?
   A. Best effort
   B. CBR
   C. VBR
   D. ABR

6. As far as the outside world is concerned, all the followings are true about NAT **EXCEPT**:
   A. Only one IP address is needed from the ISP for all devices in a network.
   B. Local administrator can change addresses of devices in his network without notifying outside world.
   C. Local administrator can change ISP without changing addresses of devices in local network.
   D. All local addresses are visible by outside world.

7. Referring to Figure 1. Use Bellman-Ford equation to determine the least cost path from node ***u*** to node ***w*** .

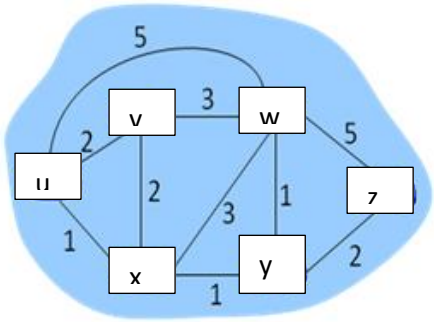


Figure1

   A. 1
   B. 2
   C. 3
   D. 5

8. These services are provided by the link layer **EXCEPT:**
   A. Error Detection
   B. Reliable delivery of data
   C. Flow Control
   D. Security of data

9. Channel partitioning protocol can be completed using the methods below **EXCEPT**:
   A. Time
   B. Space
   C. Frequency
   D. Code

10. Which of the following statements is **FALSE** about the operation of slotted ALOHA.
    A. When the node has a new frame to send, it waits until the beginning of the next slot and transmits the entire frame in the slot
    B. When there is no collision and frame is successfully transmitted, there is no retransmission
    C. Nodes will refrain from transmitting whenever it sense another node transmitting
    D. When there is collision, node detects before the end of the slot and retransmits the frame.

11. What are the contents of the data field in a frame?
    A. 64 bytes.
    B. The network layer PDU
    C. The Layer 2 source address.
    D. The data directly from the application that generated the data.

12. Which statement regarding switch is **TRUE**?
    A. Switches buffer packets
    B. Switched need to be configured
    C. Hosts are aware of the presence of switches
    D. All links share the collision domain

13. These are characteristics of Port-based VLAN **EXCEPT**:
    A. Traffic isolation
    B. Dynamic membership
    C. Inefficient use of switches
    D. Forwarding between VLANS via routing

14. A more scalable approach to interconnect VLAN switches is known as _____.
    A. VLAN tagging
    B. VLAN trunking
    C. VLAN balancing
    D. VLAN mapping

15. These are the jobs of the load balancer inside the data center **EXCEPT**:

A. Receives external client request

B. Directs workload within data center

C. Route frame to the correct link

D. Returns results to external client

16. Wireless Networks can be:

A. Infrastructure based

B. Infrastructure less

C. Both A and B

D. None

17. In cellular network, users or mobiles are directly connected to _____.

A. BTS

B. MSC

C. Both A and B

D. None

18. Which of the following wireless links has the highest data rate?

A. 802.15

B. 802.11a,g

C. 802.11n

D. 802.11b

19. Which of the following statements is **FALSE** about characteristics of wireless links?

A. Increased signal strength as it propagates

B. Decreased signal strength as it propagates

C. Interference from other sources using same wireless network frequencies

D. Multipath propagation

20. A base station in a 2G cellular network is directly connected to _____.

A. Mobile Switching Centre (MSC)

B. Public Telephone Network

C. Base Station Controller (BSC)

D. Gateway MSC

PART B [70 MARKS]

QUESTION 1 [30 MARKS]

a) What is the difference between forwarding and routing a packet performed in network layer? [2m]

b) Briefly explain the three different switching methods executed inside a router. [3m]

c) Given a Maximum Transfer Unit (MTU) for router X is 1500 bytes. If the incoming datagram size is 6000 bytes;

i. Why router X must perform fragmentation process in order to route the incoming datagram of 6000 bytes? [1m]

ii. Determine the number of fragments and draw all the fragment blocks according to the following datagram format in Figure 2. [6m]

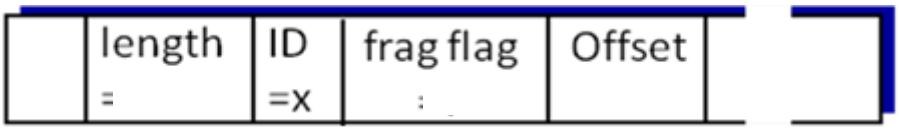


Figure 2

d) The network diagram in Figure 3 shows a classful addressing scheme. How many subnets are there in the diagram? List all the subnet addresses. Determine the class of those subnets addresses. Explain your answer. [4m]

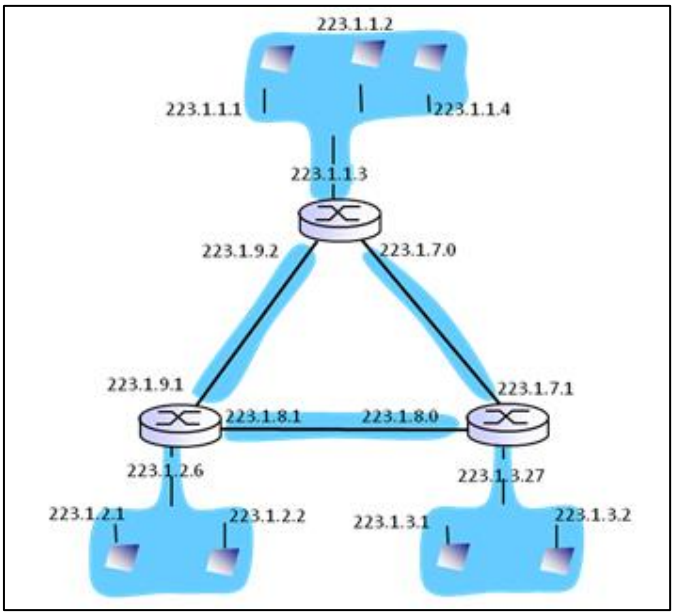


Figure 3

e) An organization needs 4500 host addresses to set up computer network.

i. How many different IP address can be assigned to users when this organization applies class B addressing scheme. [1m]

ii. How many address will be unused and how do you overcome the problems? [2m]

iii. If address format in (ii) is ***a.b.c.d/x.*** Determine the value of x which can optimize address utilization in (i)? [2m]

f) Routers A, B, C, and D have been assigned with link cost value as stated in the diagram in Figure 4. Use Dijkstra's algorithm to answer the following questions:

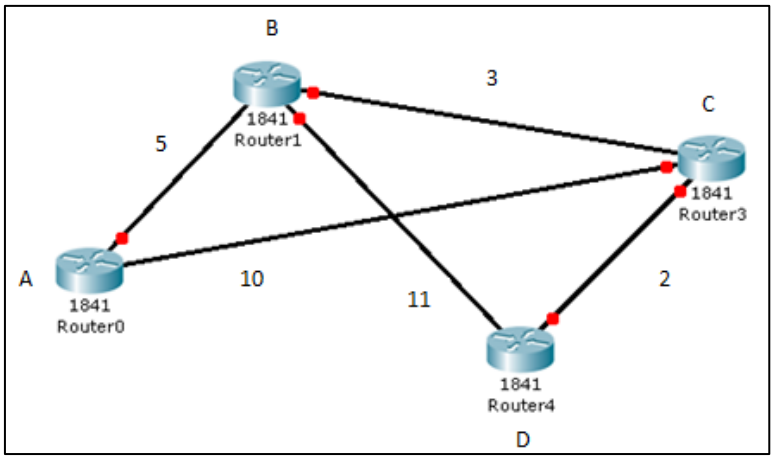


Figure 4

i. Construct the least cost path table from router D to router A. [5m]

ii. Draw the least cost paths from router D to router A. [2m]

iii. Produce the forwarding table for router D. [2m]

QUESTION 2 [30 MARKS]

a) Suppose the information content of a packet is the bit pattern 1010 1010 1011 1011 and an even parity scheme is being used. Calculate the value of the checksum field if a two-dimensional parity scheme is used. Note: each row is defined by 4 bits. [3m]

b) A packet D consists of 8-bits with the value of 10101010 is transferred over the network. The CRC error detection technique is implemented using a 4-bit Generator, G with a value of 1001.

i. Calculate the value of R. [4m]

ii. Show the value of the data that is to be transmitted. [1m]

c) Figure 5 shows three LAN interconnected by two routers and Table 1 shows the ARP table for the node at Router 1 and Router 2. Suppose Host A sends a datagram to Host E.

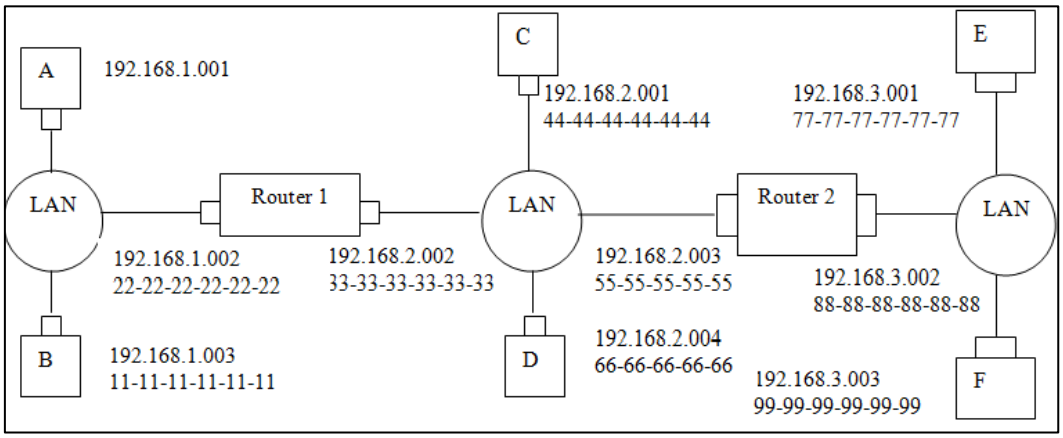


Figure 5

Table 1

IP Address	Mac Address	TTL
192.168.1.001	00-00-00-00-00-00	13:45:00
192.168.1.003	11-11-11-11-11-11	13:52:00
192.168.3.001	77-77-77-77-77-77	16:27:00
192.168.3.003	99-99-99-99-99-99	16:39:00

i. Give the source and destination IP and MAC addresses in the frame encapsulating the IP datagram as it is transmitting from A to Router 1. [2m]

ii. Give the source and destination IP and MAC address in the frame encapsulating the IP datagram as it is transmitting from Router 2 to E. [2m]

iii. Suppose the network wants to send an IP datagram from Host F to Host B. If the ARP table of Host B is empty, list the steps taken to send the IP datagram using ARP. [3m]

d) Complete Table 2 by referring to the Ethernet frame shown in Figure 6 below. [5m]

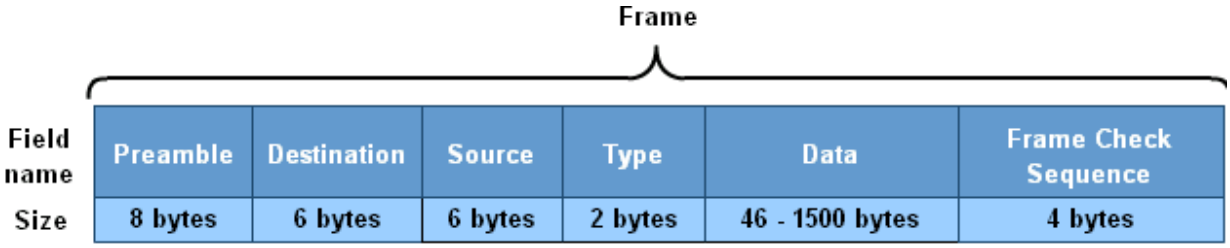


Figure 6

Table 2

Field	Explanation
	48 bit MAC address for the receiver.
	A value used to check for damaged frames.
	PDU that is to be transported over the media.
	Used for synchronization; also contains a delimiter to mark the end of the timing information.
	Value to indicate which upper layer protocol will receive the data after the Ethernet process is complete.

e) All the following questions are based on the switch operation.

i. How does switch learns which hosts can be reached through which interfaces? (i.e., name the process) [1m]

ii. Refer to Figure 7. Host 1 is sending data to Host 2. Explain the operation of the switch when the MAC address table is empty? [3m] (Answer on the next page)

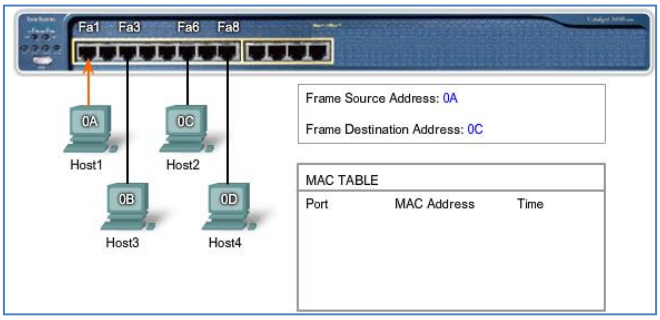


Figure 7

iii. Refer to Figure 8. Now the MAC table has a new entry. Host2 replies by sending frame to Host1. Explain the operation of the switch in this particular situation. [3m]

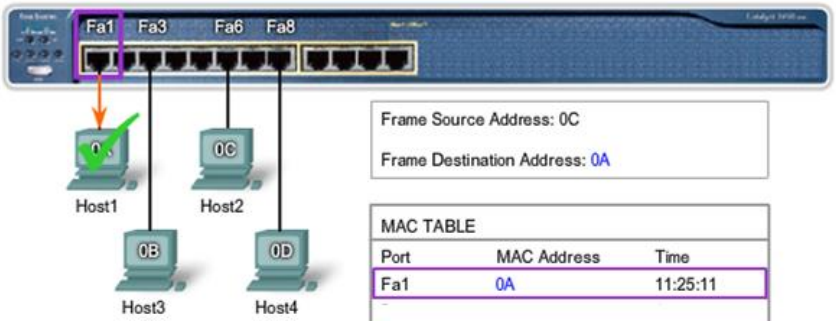


Figure 8

iv. Update the switch table in Table 3 after the latest entry. [1m]

Table 3

MAC Table		
Port	MAC Address	
Fa1	0A	11:25:11

iv. State two similarities between switches and routers. [2m]

QUESTION 3 [10 MARKS]

a) Based on the TWO elements of a wireless network, fill up Table 4 below. [3m]

Table 4

Mode of wireless network		
Main Characteristic		
One Example		

b) Suppose host A in network B is moving to network C as shown in Figure 9 below.

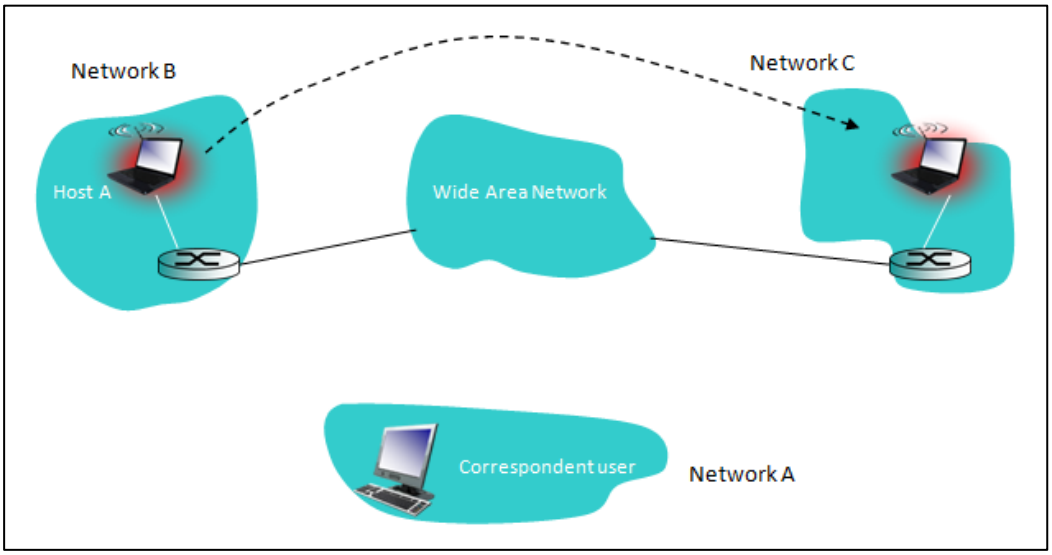


Figure 9

i. If the correspondent user is communicating with host A at new location, show the major steps that are being performed if the mobility is done via direct routing by drawing, numbering and labelling the arrow to indicate the sequence of the steps in Figure 9. [3m]

ii. Which type of host A's IP address is being used by correspondent user to forward data packets? [1m]

c) What generation of cellular architecture is shown in Figure 10? Fill in the blanks with the names of the missing elements. [3m]

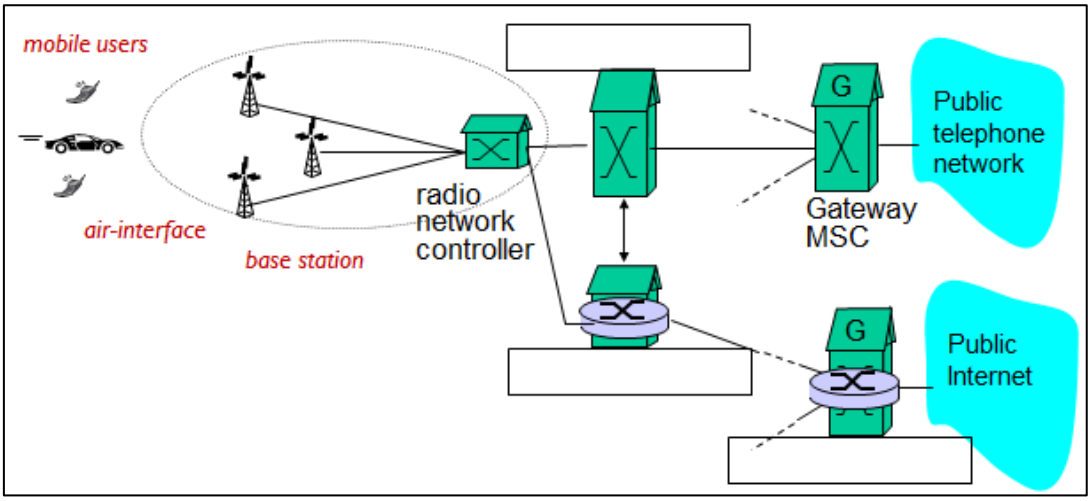


Figure 10